LMU LUDWIG-MAXIMILIANS-UNIVERSITÄT MÜNCHEN

technische universität dortmund

Advanced Statistical Learning

Chapter 2: Classification

Bernd Bischl, Julia Moosbauer, Andreas Groll

Department of Statistics – TU Dortmund

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CLASSIFICATION TASKS

In classification, we aim at predicting a discrete output

$$y \in \mathcal{Y} = \{C_1, ..., C_g\}$$

with $2 \leq g < \infty$ given data $\mathcal{D}$.

In this course, we assume the classes to be encoded as

- $\mathcal{Y} = \{0, 1\}$ or $\mathcal{Y} = \{-1, 1\}$ (in the binary case $g = 2$)
- $\mathcal{Y} = \{1, 2, ..., g\}$ (in the multiclass case $g \geq 3$)

Classifiers

CLASSIFICATION MODELS

We defined models $f : \mathcal{X} \to \mathbb{R}^g$ as functions that output (continuous) **scores** / **probabilities** and **not** (discrete) classes. Why?

- From an optimization perspective, it is **much** (!) easier to optimize costs for continuous-valued functions
- Scores / probabilities (for classes) contain more information than the class labels alone
- As we will see later, scores can easily be transformed into class labels; but class labels cannot be transformed into scores

We distinguish **scoring** and **probabilistic** classifiers.

CLASSIFICATION MODELS

Scoring Classifiers:

- Construct g **discriminant / scoring functions** $f_1, ..., f_g : \mathcal{X} \to \mathbb{R}$
- Scores $f_1(\mathbf{x}), ..., f_g(\mathbf{x})$ are transformed into classes by choosing the class with the maximum score

class itself $\in \{C_{-1}, C_1\}$

$$h(\mathbf{x}) = \underset{k \in \{1,2,...,g\}}{\arg\max} \; f_k(\mathbf{x}).$$

- For $g = 2$, a single discriminant function $f(\mathbf{x}) = f_1(\mathbf{x}) - f_{-1}(\mathbf{x})$ is sufficient (note that it would be more natural here to label the classes with $\{+1, -1\}$)

$$\begin{aligned} h(\boldsymbol{x}) &= 1 \\ \iff f_1(\mathbf{x}) &> f_{-1}(\mathbf{x}) \\ \iff f_1(\mathbf{x}) - f_{-1}(\mathbf{x}) &> 0 \\ \iff f(\mathbf{x}) &> 0 \end{aligned}$$

CLASSIFICATION MODELS

- Class labels are constructed by $h(\mathbf{x}) = \text{sgn}(f(\mathbf{x}))$
- $|f(\mathbf{x})|$ is called “confidence”

CLASSIFICATION MODELS

Probabilistic Classifiers:

- Construct g **probability functions**
  $\pi_1, ..., \pi_g : \mathcal{X} \to [0, 1],\ \sum_i \pi_i = 1$
- Probabilities $\pi_1(\mathbf{x}), ..., \pi_g(\mathbf{x})$ are transformed into labels by predicting the class with the maximum probability

$$h(\mathbf{x}) = \underset{\{k \in 1,2,...,g\}}{\arg\max}\ \pi_k(\mathbf{x})$$

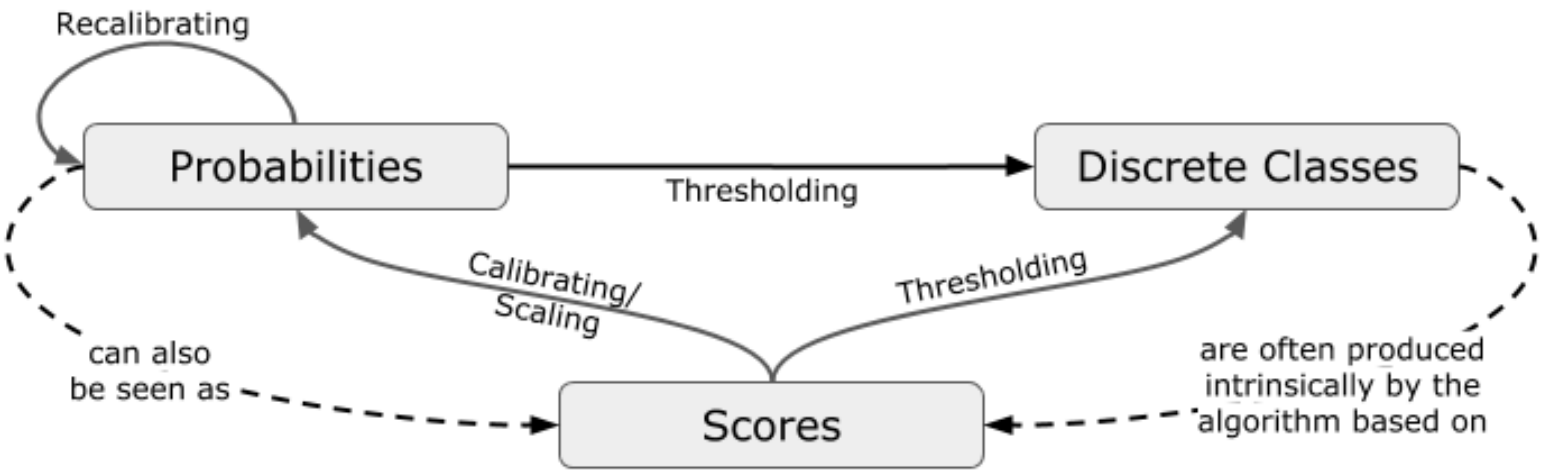
- If $g = 2$, usually one probability function $\pi(\mathbf{x})$ is outputted by most models (note that it would be more natural here to label the classes with $\{0, 1\}$) $\Rightarrow \mathbb{P}(y=1 \mid \mathbf{x}) = \mathbb{E}[y \mid \mathbf{x}] = \pi(\mathbf{x})$
- If $g = 2$, transformation into discrete classes by thresholding: $h(\mathbf{x}) := \mathbb{1}(\pi(\mathbf{x}) \geq c)$ for some threshold $c \in [0, 1]$ (for binary classification we usually set $c = 0.5$)
- Probabilistic classifiers can also be seen as scoring classifiers

CLASSIFICATION MODELS

Remark: If we want to emphasize that our model outputs probabilities we denote the model as $\pi : \mathcal{X} \to [0, 1]^g$; if we are talking about models in a general sense, we write f comprising both probabilistic and scoring classifiers (context will make this clear!)

CLASSIFICATION MODELS

- Both scoring and probabilistic classifiers can be turned into discrete classes via thresholding (binary case) / predicting the class with the maximum score (multiclass)
- Discrete classes, which are often intrinsically produced by scores, can not be transferred into scores again (Attention: Arrow in the figure might be misleading!)



DECISION BOUNDARIES

- A decision boundary is defined as a hypersurface that partitions the input space $\mathcal{X}$ into g (the number of classes) **decision regions**

$$\mathcal{X}_k = \{\mathbf{x} \in \mathcal{X} : h(\mathbf{x}) = k\}$$

- Ties between those regions are called **decision boundaries**

$$\{\mathbf{x} \in \mathcal{X} : \quad \exists\, i \neq j \text{ s.t. } f_i(\mathbf{x}) = f_j(\mathbf{x}) \text{ and } f_i(\mathbf{x}), f_j(\mathbf{x}) \geq f_k(\mathbf{x})\ \forall k \neq i, j\}$$

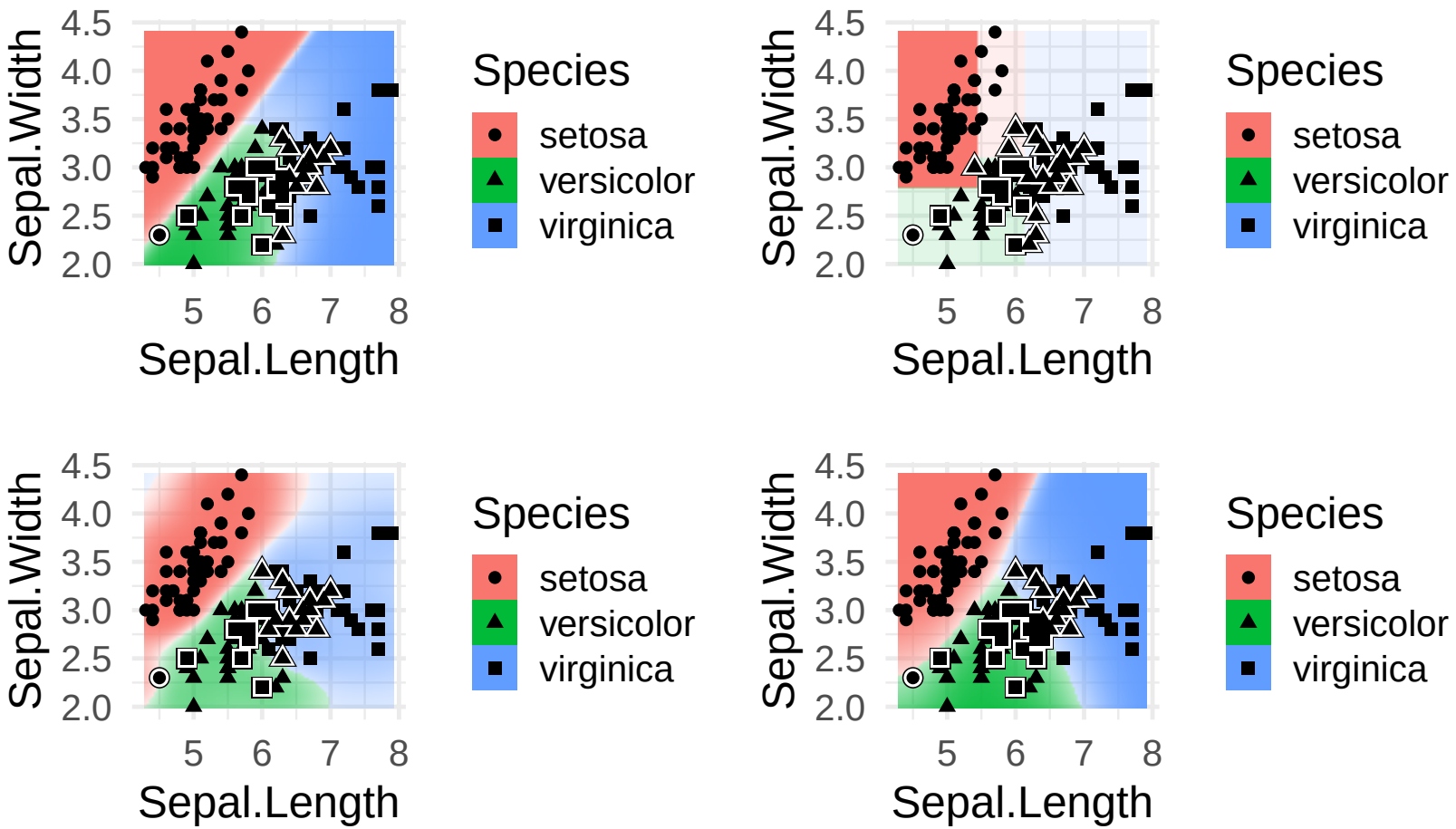
(in the general multiclass case) or

$$f(\mathbf{x}) = c\,,$$

$\{\underline{x} \in \mathcal{X} : \pi(\underline{x}) = 0.5\}$

if $c \in \mathbb{R}$ was used as threshold (usually $c = 0$ for scoring classifiers and $c = 0.5$ for probabilistic classifiers) in the binary case.

DECISION BOUNDARIES



Different shapes of decision boundaries. Classifiers (QDA, decision tree, nonlinear SVM, Naive-Bayes)

LINEAR CLASSIFIERS

If the discriminant functions $f_k(\mathbf{x})$ can be specified as linear functions (possibly through a rank-preserving, monotone transformation $g : \mathbb{R} \to \mathbb{R}$), i.e.

$$g(f_k(\mathbf{x})) = \boldsymbol{w}_k^\top \boldsymbol{x} + b_k,$$

we will call the classifier a **linear classifier**.

We can then write a tie between scores as

$$\begin{aligned} f_i(\mathbf{x}) &= f_j(\mathbf{x}) \\ \iff g(f_i(\mathbf{x})) &= g(f_j(\mathbf{x})) \\ \iff \boldsymbol{w}_i^\top \mathbf{x} + b_i &= \boldsymbol{w}_j^\top \boldsymbol{x} + b_j \\ \iff (\boldsymbol{w}_i - \boldsymbol{w}_j)^\top \mathbf{x} + (b_i - b_j) &= 0 \\ \iff \boldsymbol{w}_{ij}^\top \mathbf{x} + b_{ij} &= 0\,, \end{aligned}$$

$= : b_{ij}$, $= : \boldsymbol{w}_{ij}$, hyperplane equation

LINEAR CLASSIFIERS

with $\boldsymbol{w}_{ij} := \boldsymbol{w}_i - \boldsymbol{w}_j$ and $b_{ij} = b_i - b_j$. This is a **hyperplane** separating two classes.

Note that linear classifiers can represent non-linear decision boundaries in the original input space if we use derived features like higher order interactions, polynomial features, basis function expansions, etc.

SCALING: SIGMOIDS (BINARY)

Probabilistic classifiers are often preferred because probabilities yield a more natural interpretation. Any score-generating model can be turned into a probability estimator.

For $g = 2$, we can use a transformation function $s : \mathbb{R} \to [0, 1]$

$$\pi(\mathbf{x}) := s\left(f(\mathbf{x})\right) \in [0, 1]$$

to map scores to probabilities.

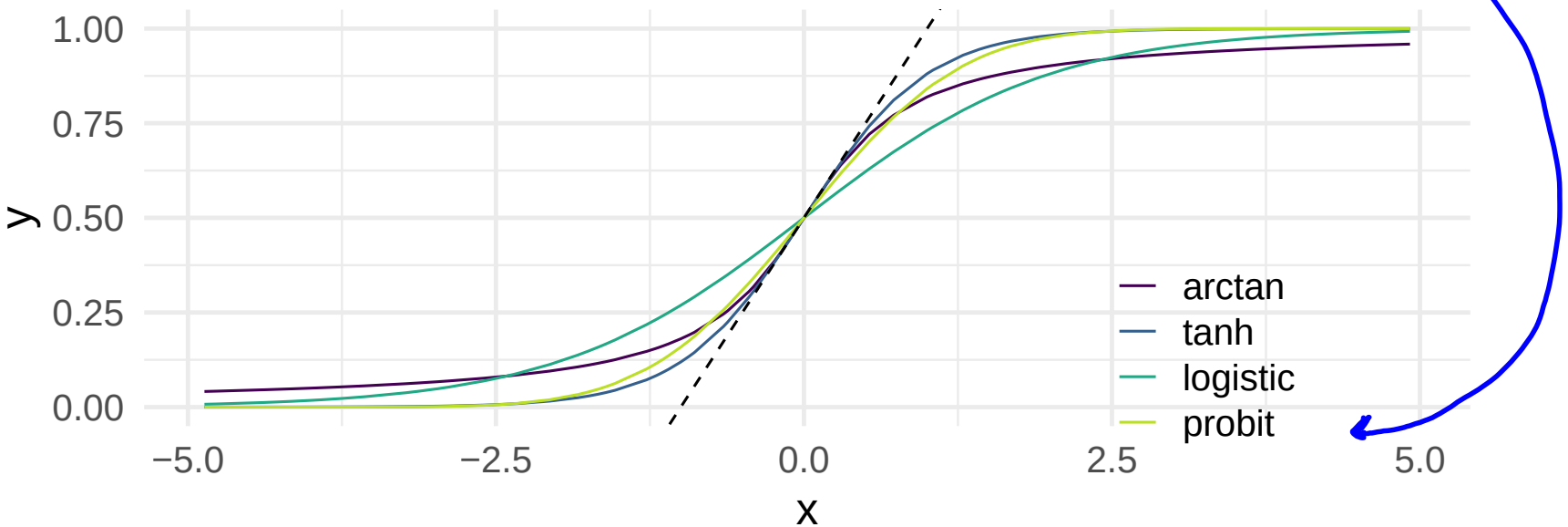
A commonly used type of transformation functions are **sigmoid** functions: A **sigmoid** is a bounded, differentiable, real-valued function $s : \mathbb{R} \to [0, 1]$ that has a non-negative derivative at each point.

In deep learning **sigmoids** are used as **activation functions**.

SCALING: SIGMOIDS (BINARY)

Examples for sigmoid functions:

- Arctan function: $s(t) = \arctan(t)$
- Hyperbolic tangent: $s(t) = \tanh(t) = \frac{e^t - e^{-t}}{e^t + e^{-t}}$
- Logistic function: $s(t) = \frac{1}{1+e^{-t}}$
- Any cumulative ~~density~~ distribution function (cdf), e.g. the normal distribution (also called probit function)



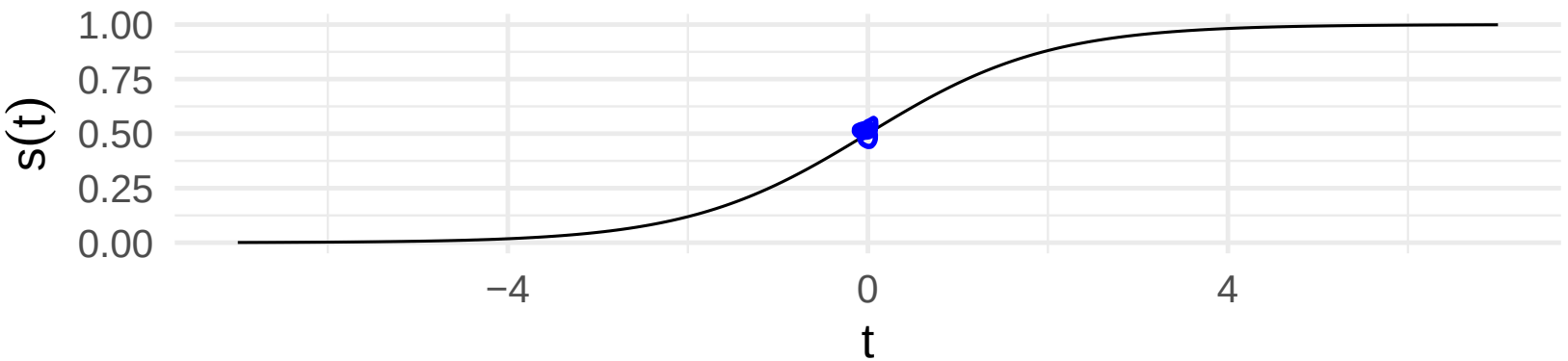
SCALING: SIGMOIDS (BINARY)

The logistic function

$$s(t) = \frac{1}{1 + e^{-t}}$$

is a popular choice for transforming scores to probabilities (used, for example, in logistic regression). Properties of the logistic function $s(t)$:

- $\lim_{t \to -\infty} s(t) = 0$ and $\lim_{t \to \infty} s(t) = 1$
- $\frac{\partial s(t)}{\partial t} = \frac{\exp(t)}{(1+\exp(t))^2} = s(t)(1 - s(t))$
- $s(t)$ is symmetrical about the point $(0, \frac{1}{2})$



SCALING: SOFTMAX (MULTICLASS)

Any multiclass scoring classifier can be transformed into probabilities using a transformation that maps the scores to a vector of probabilities

$$(f_1(\boldsymbol{x}), ..., f_g(\boldsymbol{x})) \mapsto (\pi_1(\boldsymbol{x}), ..., \pi_g(\boldsymbol{x})) \,,$$

fulfilling

- $\pi_k(\boldsymbol{x}) \in [0, 1]$ for all $k = 1, ..., g$
- $\sum_{k=1}^{g} \pi_k(\boldsymbol{x}) = 1$.

A commonly used function is the **softmax** function, which is defined on a numerical vector $\boldsymbol{z}$ (which could be our scores):

$$s(\boldsymbol{z})_k = \frac{\exp(z_k)}{\sum_j \exp(z_j)}$$

actually used: $s(z)_k = \frac{\exp(z_k)}{\sum_{j=1}^{g-1} \exp(z_j) + 1}$ $k = 1, \ldots, g-1$

SCALING: SOFTMAX (MULTICLASS)

It is a generalization of the logistic function (check for $g = 2$). It "squashes" a g-dimensional real-valued vector $\boldsymbol{z}$ to a vector of the same dimension, with every entry in the range $[0, 1]$ and all entries adding up to 1.

For a categorical response variable $y \in \{1, \ldots, g\}$ with $g > 2$ the model extends to

$$\pi_k(\mathbf{x}) = \frac{\exp(f_k(\mathbf{x}))}{\sum_{j=1}^{g} \exp(f_j(\mathbf{x}))}.$$

Compared to the **arg max** operator, the **soft max** keeps information about the other, non-maximal elements in a reversible way. Thus the function is called **soft** max.

Generative vs. Discriminative Approaches

GENERATIVE APPROACHES

Two fundamental approaches exist to construct classifiers: The **generative approach** and the **discriminant approach**.

The **generative approach** employs the Bayes theorem:

$$\pi_k(\mathbf{x}) = \mathbb{P}(y = k \mid \mathbf{x}) = \frac{\mathbb{P}(\mathbf{x}|y = k)\mathbb{P}(y = k)}{\mathbb{P}(\mathbf{x})} \propto \mathbb{P}(\mathbf{x}|y = k)\pi_k$$

and models $\mathbb{P}(\mathbf{x} \mid y = k)$ (usually by assuming something about the structure of this distribution) to allow the computation of $\pi_k(\mathbf{x})$. The discriminant functions in this approach are

$$\pi_k(\mathbf{x}) \text{ or } \log \mathbb{P}(\mathbf{x}|y = k) + \log \pi_k$$

Discriminant approaches try to model the discriminant functions directly, often by loss minimization.

GENERATIVE APPROACHES

Examples:

- Linear discriminant analysis: generative, linear. Each class's density is a **multivariate Gaussian** with

$$\mathbb{P}(\mathbf{x} \mid y = k) \sim \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma})$$

  with equal covariances.
- Quadratic discriminant analysis: generative, non-linear. Each class's density is a **multivariate Gaussian** with

$$\mathbb{P}(\mathbf{x} \mid y = k) \sim \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$

  with unequal covariances $\boldsymbol{\Sigma}_k$.

GENERATIVE APPROACHES

- Naive Bayes: generative, non-linear. "Naive" **conditional independence assumption**, i.e. features given the category y are conditionally independent of each other

$$\mathbb{P}(\mathbf{x} \mid y = k) = \mathbb{P}((x_1, ..., x_p) \mid y = k) = \prod_{j=1}^{p} \mathbb{P}(x_j \mid y = k).$$

- Logistic regression: discriminant, (usually) linear $(g = 2)$

$$\pi_1(\underline{x}) = \frac{\exp(\underline{x}^T \underline{\beta})}{1 + \exp(\underline{x}^T \underline{\beta})}$$

Multinomial logistic regression model

$$\pi_u(\underline{x}) = \frac{\exp(\underline{x}^T \underline{\beta}_u)}{\sum_{j=1}^{g-1} \exp(\underline{x}^T \underline{\beta}_j) + 1} \qquad u = 1, ..., g-1$$